Complex numbers and geometry

Lecture and worked examples

Objectives

Interpret the modulus and an argument of $\dfrac{c-a}{b-a}$; prove that points are collinear, that lines are perpendicular, the nature of a triangle; determine sets of points; determine the n-th roots of unity and use them for regular polygons.

Introductory activity

1. Factorise $z^3 - 1$ by $z - 1$, then solve $z^3 = 1$ in $\mathbb{C}$. Let j be the solution with positive imaginary part.
2. Plot the three solutions in the complex plane. Compute the distances between them. Which figure do you obtain?
3. Write j in exponential form. Check that $j^2 = \bar{j}$ and that $1 + j + j^2 = 0$. Interpret the last equality with the centroid.
4. Solve $z^4 = 1$ and plot the solutions. Same question for $z^6 = 1$, using $z^6 - 1 = \left(z^3 - 1\right)\left(z^3 + 1\right)$.

Introductory activity (2)

5. State a conjecture about the solutions of $z^n = 1$: their number, their modulus, their arguments, the figure they form.

Introductory activity (3)

Observing is not proving: the properties conjectured here are established in the course.

Angles and lengths

Let A, B, C be three points with affixes a, b, c, with $A \neq B$ and $A \neq C$. Then

$$\left|\frac{c-a}{b-a}\right| = \frac{AC}{AB} \qquad \text{and} \qquad \arg\left(\frac{c-a}{b-a}\right) \equiv \left(\overrightarrow{AB}, \overrightarrow{AC}\right) \pmod{2\pi}.$$

Angles and lengths

The modulus of a quotient is the quotient of the moduli: $|c-a| = AC$ and $|b-a| = AB$. For the argument, $\arg(b-a) \equiv \left(\vec{u}, \overrightarrow{AB}\right)$ and $\arg(c-a) \equiv \left(\vec{u}, \overrightarrow{AC}\right)$, so by Chasles' relation $\arg \dfrac{c-a}{b-a} \equiv \left(\vec{u}, \overrightarrow{AC}\right) - \left(\vec{u}, \overrightarrow{AB}\right) \equiv \left(\overrightarrow{AB}, \overrightarrow{AC}\right) \pmod{2\pi}$.

Angles and lengths

With the same notation, let $q = \dfrac{c-a}{b-a}$.

1. A, B, C are collinear if and only if q is real.
2. The lines (AB) and (AC) are perpendicular if and only if q is purely imaginary.
3. The triangle ABC is right-angled and isosceles at A if and only if $q = i$ or $q = -i$; it is equilateral if and only if $q = \mathrm{e}^{i\pi/3}$ or $q = \mathrm{e}^{-i\pi/3}$.

Angles and lengths

q is real if and only if its argument is 0 or π modulo 2π, that is if $\overrightarrow{AB}$ and $\overrightarrow{AC}$ are collinear. q is purely imaginary (nonzero) if and only if its argument is $\pm\frac{\pi}{2}$. For 3, the previous theorem gives $|q| = \frac{AC}{AB}$ and $\arg q \equiv \left(\overrightarrow{AB}, \overrightarrow{AC}\right) \pmod{2\pi}$. Hence $q = \pm i$ if and only if $AC = AB$ and $\left(\overrightarrow{AB}, \overrightarrow{AC}\right) \equiv \pm\frac{\pi}{2}$, which characterises a triangle that is right-angled and isosceles at A. Likewise, $q = \mathrm{e}^{\pm i\pi/3}$ if and only if $AC = AB$ and $\left(\overrightarrow{AB}, \overrightarrow{AC}\right) \equiv \pm\frac{\pi}{3}$. A triangle that is isosceles at A with angle A measuring $\frac{\pi}{3}$ has two other equal angles, each measuring $\frac{1}{2}\left(\pi - \frac{\pi}{3}\right) = \frac{\pi}{3}$: its three angles are equal, so it is equilateral. Conversely, an equilateral triangle satisfies $AC = AB$ and its angle at A measures $\frac{\pi}{3}$.

Angles and lengths

Let A and B have affixes a and b. The direct square $ABCD$ has vertices $c = b + i(b - a)$ and $d = a + i(b - a)$: we go from $\overrightarrow{AB}$ to $\overrightarrow{BC}$ by a direct quarter turn. With $a = 0$ and $b = 1$: $c = 1 + i$ and $d = i$.

n-th roots of unity

Let $n \geq 1$ be an integer. An n-th root of unity is any complex number z such that $z^n = 1$. Their set is written $\mathbb{U}_n$.

n-th roots of unity

For every integer $n \geq 1$, $\mathbb{U}_n$ has exactly n elements:

$$\mathbb{U}_n = \left\{ \mathrm{e}^{2ik\pi/n} \ : \ k \in \{0, \ldots, n-1\} \right\}.$$

n-th roots of unity

Let $z \in \mathbb{U}_n$. Then $|z|^n = |z^n| = 1$ with $|z| \geq 0$, so $|z| = 1$ and $z = \mathrm{e}^{i\theta}$ for some real number θ. The equality $\mathrm{e}^{in\theta} = 1$ is equivalent to $n\theta \equiv 0 \pmod{2\pi}$, that is $\theta = \dfrac{2k\pi}{n}$ with k an integer. Conversely, $\left(\mathrm{e}^{2ik\pi/n}\right)^n = \mathrm{e}^{2ik\pi} = 1$.

n-th roots of unity (2)

Two integers k and k' give the same number if and only if $\frac{2(k-k')\pi}{n}$ is a multiple of 2π, that is $k \equiv k' \pmod{n}$. Every integer k has the same remainder as an integer of $\{0, \ldots, n-1\}$, and these n integers have distinct remainders: $\mathbb{U}_n$ has exactly n elements.

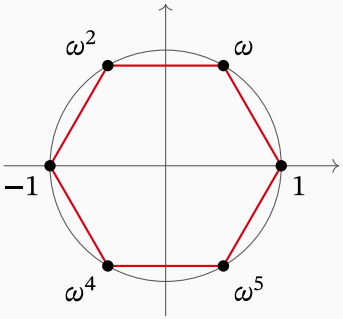
n-th roots of unity

For $n \geq 3$, the images of the elements of $\mathbb{U}_n$ are the vertices of a regular polygon with n sides inscribed in the unit circle, one of whose vertices is the point with affix 1. With $\omega = \mathrm{e}^{2i\pi/n}$, we have $\mathbb{U}_n = \{1, \omega, \omega^2, \ldots, \omega^{n-1}\}$. Cases to know: $\mathbb{U}_2 = \{1, -1\}$, $\mathbb{U}_3 = \{1, j, j^2\}$ with $j = \mathrm{e}^{2i\pi/3}$, $\mathbb{U}_4 = \{1, i, -1, -i\}$.

n-th roots of unity

For every integer $n \geq 2$, the sum of the n-th roots of unity is zero.

n-th roots of unity

With $\omega = e^{2i\pi/n} \neq 1$ (since $n \geq 2$):

$$1 + \omega + \cdots + \omega^{n-1} = \frac{1 - \omega^n}{1 - \omega} = 0.$$